

Functional Requirement - Manhattan

Software Requirement

Item	Isi
Nama	Manhattan
Studi Kasus	Aplikasi real-time group chat dengan end-to-end encryption.
Target Pengguna	Pengguna yang membutuhkan ruang chat sementara, cepat, dan privat melalui browser.
Permasalahan	Chat biasa sering menyimpan plaintext di server, sulit menjaga privasi, dan rawan bentrok sesi ketika b
Solusi	Manhattan memakai WebSocket/STOMP untuk komunikasi real-time, Web Crypto API untuk RSA/AES

Functional Requirement

ID	Functional Requirement
FR-01	User dapat membuat room baru dengan nama alphanumeric 3-15 karakter.
FR-02	User dapat bergabung ke room yang sudah ada dengan password jika room diproteksi.
FR-03	Sistem memvalidasi satu sesi aktif per IP address pada WebSocket handshake.
FR-04	Client mengenkripsi pesan dengan AES-256-GCM sebelum dikirim ke server.
FR-05	Client bertukar AES key secara aman menggunakan RSA public key antar peserta.
FR-06	Server meneruskan ciphertext ke topic room tanpa membaca plaintext pesan.
FR-07	Sistem membatasi percobaan password salah sebanyak 5 kali lalu logout 60 detik.
FR-08	User dapat melihat daftar peserta aktif di room.

Screenshot Kode

Requirement Evidence - Client RoomController

```
39      * Validate a room name against the naming rules.
40      * Rules:
41      *   - Must be 3-15 characters in length
42      *   - Only alphanumeric characters (a-z, A-Z, 0-9)
43      *   - No spaces
44      *   - Case-sensitive (validation does not normalize case)
45      *
46      * @param {string} name - The room name to validate
47      * @returns {{ valid: boolean, error: string|null }}
48      */
49      validateRoomName(name) {
50          if (name === null || name === undefined) {
51              return { valid: false, error: 'Room name is required' };
52          }
53
54          if (typeof name !== 'string') {
55              return { valid: false, error: 'Room name must be a string' };
56          }
57
58          if (name.length < 3) {
59              return { valid: false, error: 'Room name must be at least 3 characters' };
60          }
61
62          if (name.length > 15) {
63              return { valid: false, error: 'Room name must be at most 15 characters' };
64          }
65
66          if (/\/s/.test(name)) {
67              return { valid: false, error: 'Room name must not contain spaces' };
68          }
69
70          if (!/^[a-zA-Z0-9]+$/.test(name)) {
71              return { valid: false, error: 'Room name must contain only alphanumeric characters' };
72          }
73
74          return { valid: true, error: null };
75      }
76
77      /**
78       * Subscribe to the private queue for receiving server responses.
79       * Should be called after WebSocket connection is established.
80       */
81      subscribeToPrivateQueue() {
82          if (this._privateSubscription) {
83              return; // Already subscribed
84          }
85
86          this._privateSubscription = this._wsClient.subscribe(
87              '/user/queue/private',
88              (message) => {
89                  this._handleServerMessage(message);
90              }
91          );
92      }
93  }
```